

Andrés Herencia

Forward Deployed AI Engineer

GenAI • MLOps • Cloud architecture

InnoIT @ Moody's, Madrid

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About me

Profile

Telecommunication Engineer from the [Polytechnic University of Madrid](#) (top Spanish university in Telecommunication Engineering) and Master's Degree in Data Science, with [over four years](#) of experience designing and deploying Generative AI solutions, MLOps pipelines, and high-performance, scalable cloud architectures.

Identity

Forward Deployed AI Engineer
InnoIT @ Moody's • Madrid

Credentials

2× AWS-Certified:
Solutions Architect (SAA) &
Machine Learning Engineer
(MLA)

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Professional experience

Career path

Period	Role	Organisation
Jan 2026 – now	AI Engineer	Moody's (InnoIT)
Sep 2024 – Jan 2026	ML Engineer	BBVA (Minsait)
Feb 2024 – Sep 2024	DevOps / MLOps	The Cliff
Oct 2022 – Apr 2023	Big Data researcher	UPM / SUNRISE
Dec 2021 – Jun 2022	Data Scientist	Cabify Chair

Since January 2026

Madrid, Spain

Building MCP tools and agents for interactive financial workflows (PESTLE reports, executive summaries, and related products) on the Moody's GenAI Assistant.

- New assistant features, event-driven productisation, and QA
- Reviews and engineering standards for collaborators
- Internal use cases for consulting engagements and technical interviews

Moody's

Data, intelligence and analytical tools so business and financial leaders can make confident decisions — more than 115 years of expertise.

moodys.com

Moody's · selected deliveries

- Ready-to-use [agentic workflows](#), including an HMAS (Hierarchical Multi-Agent System) that builds automatic financial memories for Moody's partner companies.
- Agentic product that generates [EWS reports](#) automatically.
- Stack: TypeScript, React, Node.js, AWS, GitHub, OpenAI, Gemini, Claude.

Flagship result

Agentic EWS reports: **5–20 days** → **3–15 min** (~99%+ faster).

BBVA (Minsait) · Machine Learning Engineer

September 2024 – January 2026

1 year 5 months · Madrid, Spain

Industrialising Generative AI systems and web applications for BBVA, in hybrid environments: cloud-native (AWS & Azure) and on-premises.

- Productivise GenAI projects that support banking processes.
- Design fully [serverless](#) architectures on AWS.
- Enforce best practices for Generative AI applications.

BBVA

Global financial group founded in 1857. Present in 25+ countries; strong leadership in Spain, Mexico, South America and Turkey.

bbva.com

BigRAG

Hybrid RAG architecture for large-scale data ingestion on AWS.

LLM Evaluator

Automatic evaluation of LLMs and RAG systems on Azure.

BBVA Claims

Email + attachment classification:

- 30–60 min → **< 3 min**
- Quality metrics up to **90–95%**
- Cost **< 50 cents** per mail

Python · AWS · Azure · BitBucket · LangChain · OpenAI

The Cliff · DevOps / MLOps Engineer

February 2024 – September 2024

8 months · Madrid, Spain

Cloud & AI on **Maind** (enterprise LLM platform) and **Logos360** (HR evaluation).

MLOps	Data science	DevOps
Observability with Langfuse.	RAG on Qdrant.	ECS, EKS, EC2, GitLab CI/CD.

The Cliff

Spanish start-up and consultancy that ships pioneer AI-based products.

the-cliff.com

October 2022 – April 2023

7 months · Madrid, Spain

R&D engineer on the European [SUNRISE](#) project: resilience of critical infrastructures through data analysis and optimisation.

- Geospatial processing with the Google Earth API.
- Mathematical optimisation of transport availability for the [CRTM](#) metro network.
- Analytical assessment of water distribution networks (WDN) to support crisis resilience for ACOSOL (Costa del Sol).

SUNRISE

EU project on crisis resilience for critical infrastructures, including potable water (ACOSOL) and public transport (CRTM).

sunrise-europe.eu

December 2021 – June 2022

7 months · Madrid, Spain

Machine learning modules for a multi-agent simulation used to test marketplace pricing safely, outside production.

- Agent generation with time-series models ([SARIMA](#)) and Gaussian Mixture Models ([GMM](#)).
- Transactional behaviour with multimodal logistic regression.
- Simulator still used internally to refine marketplace strategy.

Cabify

Ibero-American mobility platform (ride-hailing, corporate transport, logistics) in 40+ cities across 6 countries.

cabify.com

Featured work

- **SplitterMR** — Python library for LLM-ready document chunking ([PyPI](#), 20+ GitHub stars, used internally at Minsait).
- **PresentIA** — MVP that turns prompts and media into editable PowerPoint decks (HORSE Powertrain / Renault Group, tenant InnoIT).
- **AWS CLI MCP** — agent that runs AWS CLI instructions over the Model Context Protocol.
- **Master's thesis** — mathematical analysis of the Transformer and an LLM for mental health counselling.

Library with four building blocks: **Readers**, **Models & embedders**, and **Splitters**.

- Readers extract file content to Markdown (Docling, MarkItDown, or a custom base class).
- Splitters chunk by characters, grammar, tokens, code separators, semantic similarity, or schema.
- Visual Language Models read scans, images and tables (OpenAI, Anthropic, Hugging Face, Gemini, Grok, or BYO model).

Status

Published on PyPI • 20+ GitHub stars • used on internal Minsait projects

github.com/andreshere00/Splitter_MR

PresentIA (Shadowfax)

End-to-end MVP for HORSE Powertrain (Renault Group), tenant InnoIT: prompts and user media become editable **.pptx** decks.

- FastAPI: auth, projects, uploads, generation, storage
- Next.js App Router UI with shadcn/ui
- Vertex AI + python-pptx on GCS templates
- Terraform IaC: Cloud Run, Firestore, GCS (with ADRs)
- Path to InnoIT (Render + Supabase) via deployment profiles

Working MVP

Create project, upload assets, generate and download editable decks. Cost in EUR stored in Firestore. 3/3 E2E gate on Cloud Run. Workflow: **mise**, Ruff, ESLint, pytest.

Python, FastAPI, Pydantic, GCP, TypeScript, React, Docker, Terraform, Supabase

Other builds

GenAI application template

Cookiecutter for Python LLM backends: providers, services and infrastructure chosen up front.

github.com/andreshere00/GenAI-app-template

Fantasy MCP CLI

TypeScript (ESM) LaLiga Fantasy engine with Clean Architecture. MCP tools (search + scrape) help an LLM pick the weekly lineup.

github.com/andreshere00/fantasy_mcp

AWS MCP application

DDD / hexagonal MCP server that executes AWS CLI via an LLM agent (work in progress, multi-model).

github.com/andreshere00/aws_cli_mcp

Education

2024 · Master's Degree

Statistical and Computational Data Science

Universidad Complutense de Madrid (with Universidad Politécnica de Madrid)

2022 · Bachelor's Degree

Telecommunication Engineering

Universidad Politécnica de Madrid

Master's thesis

Mathematical analysis of the Transformer architecture and application on a large language model for mental health counselling.

[Read the thesis](#)

Certifications & skills

Solutions Architect – Associate

April 2025

Secure, well-architected distributed systems: scalable, resilient, efficient, fault-tolerant.

[View badge](#)

ML Engineer – Associate

November 2024

Ingest, transform, train, deploy and monitor ML workloads in production.

[View badge](#)

Courses

Course	Date
Mastering Software Architecture Patterns and System Design	May 2026
AI Engineer Agentic Track: Agents and MCP	November 2025
MLflow in Action	October 2025
FastAPI — The Complete Course	July 2025
A deep understanding of deep learning	April 2024

Skills by expertise

Advanced ●●●

Proficient ●●●

Working ●●●

Data science

GenAI & LLMs ●●●

Tools & agents ●●●

MCP ●●●

Machine Learning ●●●

Statistical methods ●●●

Deep Learning ●●●

Software

Python ●●●

Git ●●●

JavaScript & TS ●●●

Design patterns ●●●

Good practices ●●●

Java ●●●

Cloud

Solutions arch. ●●●

ML services ●●●

Compute & storage ●●●

CI/CD ●●●

IaC ●●●

Languages: Spanish C2 (native) · English C1 (full work proficiency) · Italian (learning)

Conferences and events

AWS Summit Madrid 2026 · participant

One-day event on AI and cloud: agentic AI, serverless, and how those technologies are reshaping industry.

Applied AI beyond the hype (2026) · participant

- *AI Grading* — Estíbaliz Torralbo: AI-assisted quality assessments that reduce uncertainty when evaluating devices.
- *Behind the GenAI Buzzwords* — Alessandro Canino: embeddings, retrieval, agents, evaluation, observability, hallucination control, and security.

Conclusion

Summary

I design and ship [Generative AI products](#) — agents, RAG systems, and MLOps — on AWS, Azure and GCP.

- Forward Deployed AI Engineer at **InnoIT @ Moody's**
- Proven delivery in banking (BBVA), industry (HORSE / Renault), mobility (Cabify) and European R&D (SUNRISE)
- 2× AWS-Certified (SAA + MLA)
- Open-source maintainer of [SplitterMR](#)

github.com/andreshere00 • [LinkedIn](#) • [Email](#)

Let's talk.

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